SKITII TRIALS PLATFORM

18-Day Task List

Day-by-day delivery plan for Database Schema, Mobile App Backend APIs, and Trial Admin Panel Backend APIs (Days 1–18 of 36).

VERSION
1.1

DATE
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CUSTOMER
Yenepoya Hospital

SCOPE
Trial Backend (Mobile App APIs + Trial Admin Panel APIs) + Database Schema

Goal for Day 18

By end of Day 18:

- Database schema complete
- All **Mobile App Backend APIs** complete
- All **Trial Admin Panel Backend APIs** complete (Trials, Patients, Sessions, Weekly Forms, Statistics, Institutes, PIN, Export)
- Admin Panel Frontend already started (from Day 15) with core modules wired

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Overview

Days	Workstream	Duration
1 – 3	Database Schema Design	3 days
3 – 12	Backend APIs — Mobile App	9 days
8 – 17	Backend APIs — Trial Admin Panel (parallel)	10 days
15 – 18	Admin Panel Frontend kickoff + API polish	overlaps

> Admin Panel Backend starts on Day 8 and runs through Day 17. Frontend starts Day 15 once Trials / Patients / Sessions APIs are ready.

Days 1 – 3 — Database Schema Design

Day	Tasks	Status
Day 1	Design & implement core collections: `trials`, `trialConfigurations`, `patients`, `sessions` — indexes, field types, `institutionId` / `trialId` scoping	☐
Day 2	Form collections: `preTherapy`, `postTherapy`, `sessionReadings`, `weeklyForms`, `weeklyFormPins`, `missedSessionLogs` (stats = compute-on-read; no `trialStatistics` collection)	☐
Day 3	Finalize indexes, team review. Product DB: add `controlPatientMusic` on `music_assets`. Begin Mobile App Backend APIs.	☐

Deliverable: All MongoDB collections ready for API development.

Days 3 – 12 — Backend APIs (Mobile App)

Day	Tasks	Status
Day 3	Device PIN APIs — create PIN, verify PIN, forgot PIN / OTP reset	☐
Day 4	Patient APIs — enroll (duplicate `registrationNo` when provided, dynamic schema), list, view, update (Week 0 lock)	☐
Day 5	`POST /sessions/start` — shift window, shift ordering, duplicate session, 30-min minimum	☐
Day 6	`POST /sessions/validate`; stop (`INCOMPLETE` / `USER_ENDED`) / complete (`COMPLETED` / `SESSION_MAX_REACHED`) / readings; 20-min auto-end + 30-min timeout (`AUTO_TERMINATED` / `TIMEOUT`)	☐
Day 7	Pre Therapy + Post Therapy submit APIs; `TRIAL_COMPLETED` enforcement	☐
Day 8	Weekly Form PIN verify + dynamic even-week steps submit (non-HRV). **Admin Panel Backend starts (Trials module).**	☐
Day 9	Weekly form 5-min HRV (when configured); Week 0 enrollment-day + Shift 1 30-min rule; sequential even-week ordering	☐
Day 10	Mobile statistics API (date range); shift window + ordering enforcement polish	☐
Day 11	Cron jobs — 14-day PIN rotation; 4 AM missed session email; form reminder on **last** Thu/Fri/Sat before window expiry	☐
Day 12	Mobile API polish — edge cases, error codes, config fetch. **Mobile Backend complete.**	☐

Deliverable: All Mobile App Backend APIs ready for React Native integration.

Days 8 – 17 — Backend APIs (Trial Admin Panel)

Runs in parallel with Mobile App APIs from Day 8. Continues through Day 17.

Module plan

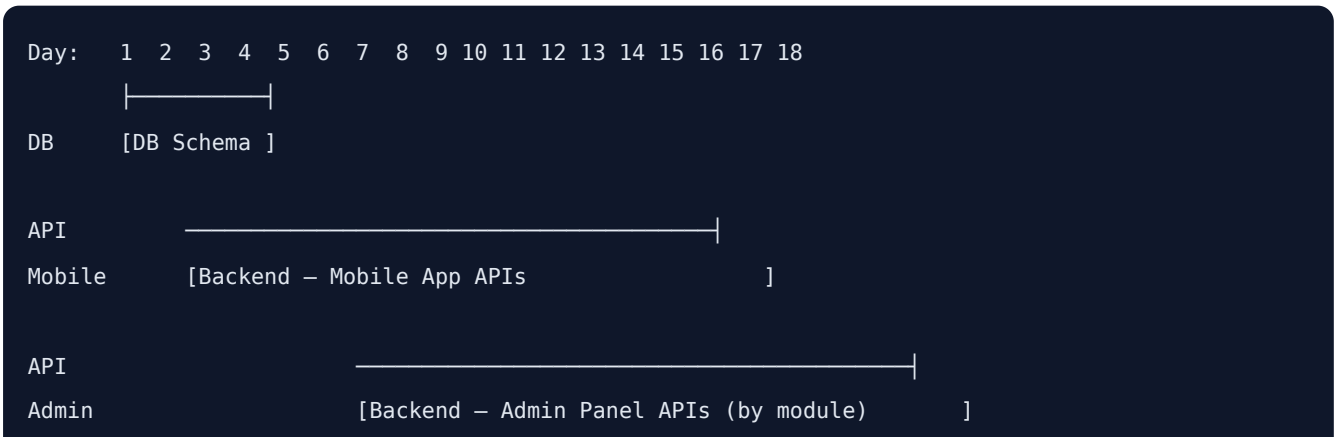
Module	Scope
Trials	Create / update / list / archive trial; patient schema; shift windows; reminders; activate/deactivate
Trial config	Assign institute → trial; admin/institute email recipients
Patients	Admin list, filters, detail, export (CSV/PDF); Week 0 view-only rules
Sessions	History, detail, Pre/Post Therapy, readings; filter by status / terminationReason
Weekly forms	List/view configured steps for that week; PIN view (read-only); missed weekly filters
Statistics	Admin KPIs, missed session logs, analytics export
Institutes & music	Institute–trial linking; `controlPatientMusic` flag APIs

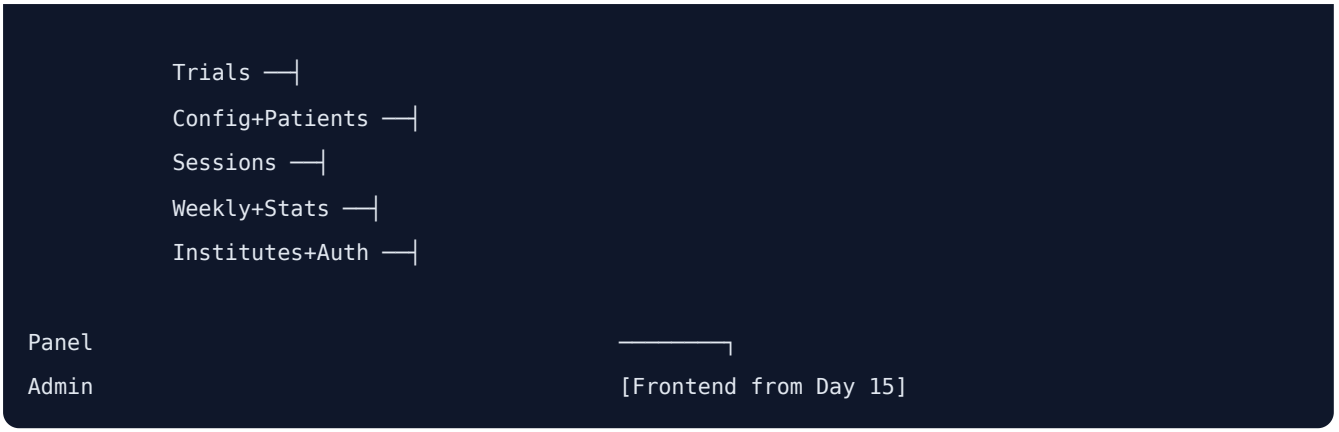
Day-by-day

Day	Tasks	Status
Day 8	**Trials module** — create / update / list trial; seed Yenepoya v1 config	☐
Day 9	**Trials module** — `patientSchema` + `weeklyFormDefinitions` CRUD (form weeks independent of `trialDurationWeeks`); shift windows; reminder days; activate/deactivate	☐
Day 10	**Trial config** — institute ↔ trial assignment; email recipients. **Patients module** — admin list + filters	☐
Day 11	**Patients module** — detail view, compliance summary, export CSV/PDF	☐
Day 12	**Sessions module** — history list (patient / date / shift); filters by status	☐
Day 13	**Sessions module** — session detail, Pre Therapy, Post Therapy, readings view/download	☐
Day 14	**Weekly forms module** — list/view dynamic steps for even weeks; missed filters; Weekly PIN view API	☐
Day 15	**Statistics module** — admin KPIs, missed session logs. **Admin Panel Frontend starts.**	☐
Day 16	**Institutes module** — Auth institution linking. Music flag APIs (`controlPatientMusic`)	☐
Day 17	Admin auth guards (Skitii internal only); Admin API review + integration smoke with frontend	☐
Day 18	Admin API polish — pagination, error codes, export edge cases; handoff checklist for frontend	☐

Deliverable: All Trial Admin Panel Backend APIs complete and ready for full frontend integration.

Parallel Work View (Days 1–18)





What Comes After Day 18 (Reference)

Days	Work
15 – 22	Trial Admin Panel Frontend (continues)
15 – 17	Skitii Admin Panel Update (`controlPatientMusic` UI)
23 – 27	Deployment (UAT + Production)
26 – 36	Testing & Bug Fixes

Full project duration: 36 working days.

Checkpoint Summary for Management

Checkpoint	Day	Expected Outcome
Schema ready	Day 3	All collections designed and implemented
Core mobile session APIs ready	Day 7	Patient + Session start/stop/complete + forms working
Weekly + cron ready	Day 11	Weekly forms, PIN, and cron jobs working
Mobile APIs complete	Day 12	Full mobile backend ready for app team
Trials + Patients + Sessions admin APIs ready	Day 13	Frontend can start on core modules
Admin APIs complete	Day 17–18	Full admin backend ready for remaining frontend modules

Risks for First 18 Days

Risk	Mitigation
Dynamic patient schema complexity	Finalize schema in Trials module (Days 8–9) before Patients admin APIs
Admin API scope under-estimated	Explicit module breakdown (Trials / Patients / Sessions / Weekly / Stats / Institutes)
Frontend blocked waiting for APIs	Frontend starts Day 15 after Sessions module lands
Parallel Admin + Mobile conflicts	Shared services review on Day 17
Cron / email delivery issues	Test email templates on Day 11